pyResToolbox MCP

Formula Reference Guide

Quick Reference for Reservoir Engineering Calculations

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March 11, 2026

Version 2.1

Abstract

This document provides a comprehensive reference guide for calculation tools available in the pyResToolbox Model Context Protocol (MCP) server. It includes mathematical formulas, symbol definitions, unit systems, and correlation methods covering oil PVT, gas PVT, well performance, simulation support, brine properties, reservoir heterogeneity analysis, and geomechanics/wellbore stability. All calculations use Field Units (US Oilfield standard).

Based on pyResToolbox: This MCP server is built upon the excellent *pyResToolbox* library by Mark W. Burgoyne (https://github.com/mwburgoyne/pyResToolbox), which provides the underlying reservoir engineering calculations.

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1 Introduction

1.1 About pyResToolbox MCP

pyResToolbox MCP is a Model Context Protocol server that provides AI agents like Claude with access to industry-standard petroleum engineering calculations. Built on the comprehensive pyResToolbox library by Mark Burgoyne, it offers 108 production-ready tools for reservoir engineering workflows. Supports both Field Units (psia, °F, ft) and Metric Units (barsa, °C, m) via the `metric` parameter.

1.2 Unit System

All calculations use Field Units (US Oilfield standard):

Property	**Unit**	**Example**
Pressure	psia	3000 psia
Temperature	°F	180°F
Permeability	mD	100 mD
Pay Thickness	ft	50 ft
Viscosity	cP	0.85 cP
Oil Rate	STB/day	542 STB/day
Gas Rate	MSCF/day	1250 MSCF/day
Oil Gravity	API° or SG	35° API
Gas Gravity	SG (air=1)	0.75
Solution GOR	scf/stb	800 scf/stb
Oil FVF	rb/stb	1.25 rb/stb
Gas FVF	rcf/scf	0.0045 rcf/scf
Compressibility	1/psi	1.2×10^{-5} 1/psi
Density	lb/ft^3	42.5 lb/ft^3

Table 1: Field Units Used in All Calculations

2 Oil PVT Calculations

2.1 Bubble Point Pressure (P_b)

2.1.1 Description

The bubble point pressure is the pressure at which the first gas bubble evolves from solution oil. It is a critical PVT property that defines the boundary between undersaturated (single-phase) and saturated (two-phase) oil behavior.

2.1.2 Available Methods

1. Standing (1947) - STAN

$$P_b = 18.2 \left[\left(\frac{R_{sb}}{\gamma_g} \right)^{0.83} \times 10^a - 1.4 \right] \tag{1}$$

where:

$$a = 0.00091 \times T - 0.0125 \times \text{API} \tag{2}$$

Symbol Definitions:

- P_b: Bubble point pressure (psia)
- R_{sb}: Solution GOR at bubble point (scf/stb)
- γ_g: Gas specific gravity (air=1)
- T: Temperature (°F)
- **API**: Oil API gravity (degrees)

2. Valko-McCain (2003) - VALMC

$$P_b = -C_1 - C_2 \times A - C_3 \times A^2 \tag{3}$$

where:

$$A = \ln \left(\frac{R_{sb}}{\gamma_{sp}} \right) \tag{4}$$

$$C_1 = 7.916 \times 10^{-4} \times \text{API}^{1.541} - 4.561 \times 10^{-5} \times T^{1.3911} \tag{5}$$

$$C_2 = 4.51 \times 10^{-5} \times \text{API}^{1.3911} \times T^{0.7849} \tag{6}$$

$$C_3 = 1.046 \times 10^{-7} \times \text{API}^{1.3911} \times T^{1.007} \tag{7}$$

Additional Symbols:

- γ_{sp}: Separator gas specific gravity (air=1)
- C_1, C_2, C_3: Correlation coefficients

3. Velarde-Blasingame-McCain (1997) - VELAR

$$\ln(P_b) = a_0 + a_1 \ln(R_{sb}) + a_2[\ln(R_{sb})]^2 + a_3 \ln(\gamma_{sp}) + a_4 \ln(\text{API}) + a_5 \ln(T + 460) \quad (8)$$

Coefficients a_0 through a_5 are empirically determined from regression analysis.

2.2 Solution Gas-Oil Ratio (R_s)

2.2.1 Description

The solution gas-oil ratio represents the volume of gas dissolved in oil at a given pressure and temperature. It increases with pressure up to the bubble point, then remains constant (equal to R_{sb}) above the bubble point.

2.2.2 Standing (1947) Method

$$R_s = \gamma_g \left[\frac{P}{18.2} + 1.4 \right] \times 10^{-a} \quad (9)$$

where:

$$a = 0.0125 \times \text{API} - 0.00091 \times T \quad (10)$$

Symbol Definitions:

- R_s: Solution gas-oil ratio (scf/stb)
- P: Pressure (psia)
- γ_g: Gas specific gravity (air=1)
- **API**: Oil API gravity (degrees)
- T: Temperature (°F)

Behavior:

- $P < P_b$: R_s calculated from correlation
- $P \geq P_b$: $R_s = R_{sb}$ (constant)

2.3 Oil Formation Volume Factor (B_o)

2.3.1 Description

The formation volume factor relates the volume of oil at reservoir conditions to its volume at stock tank conditions. It accounts for thermal expansion and gas in solution.

2.3.2 McCain et al. (1988) Method - MCAIN

$$B_o = \frac{\rho_{ST}}{\rho_o} \quad (11)$$

where oil density ρ_o is calculated from:

$$\rho_o = \frac{\rho_{ST} + 0.01357 \times R_s \times \gamma_g}{0.972 + 0.000147 \times (T + 460)^{1.175}} \quad (12)$$

Symbol Definitions:

- B_o: Oil formation volume factor (rb/stb)
- ρ_o: Oil density at reservoir conditions (lb/ft^3)
- ρ_{ST}: Oil density at stock tank conditions (lb/ft^3)
- R_s: Solution gas-oil ratio (scf/stb)
- γ_g: Gas specific gravity (air=1)
- T: Temperature (°F)

2.3.3 Standing (1947) Method - STAN

$$B_o = 0.9759 + 0.00012 \left[R_s \sqrt{\frac{\gamma_g}{\gamma_o}} + 1.25T \right]^{1.2} \tag{13}$$

Additional Symbols:

- γ_o: Oil specific gravity (water=1)

2.4 Oil Viscosity (μ_o)

2.4.1 Description

Oil viscosity is a measure of the oil's resistance to flow. It decreases with increasing temperature and decreasing pressure (above bubble point).

2.4.2 Beggs-Robinson (1975) Method - BR

Dead Oil Viscosity:

$$\mu_{od} = 10^X - 1 \tag{14}$$

where:

$$X = T^{-1.163} \times \exp(6.9824 - 0.04658 \times \text{API}) \tag{15}$$

Live Oil Viscosity:

$$\mu_o = A \times \mu_{od}^B \tag{16}$$

where:

$$A = 10.715 \times (R_s + 100)^{-0.515} \tag{17}$$

$$B = 5.44 \times (R_s + 150)^{-0.338} \tag{18}$$

Symbol Definitions:

- μ_o: Oil viscosity (cP)
- μ_{od}: Dead oil viscosity (cP)

- R_s: Solution gas-oil ratio (scf/stb)
- T: Temperature (°F)
- **API**: Oil API gravity (degrees)

2.5 Oil Density (ρ_o)

2.5.1 Description

Oil density at reservoir conditions accounts for dissolved gas and thermal expansion effects.

2.5.2 Standing-White-McCain-Hill (1995) Method - SWMH

$$\rho_o = \frac{62.4 \times \gamma_o + 0.0136 \times R_s \times \gamma_g}{B_o} \tag{19}$$

Symbol Definitions:

- ρ_o: Oil density (lb/ft^3)
- γ_o: Oil specific gravity (water=1)
- γ_g: Gas specific gravity (air=1)
- R_s: Solution gas-oil ratio (scf/stb)
- B_o: Oil formation volume factor (rb/stb)

2.6 Oil Compressibility (c_o)

2.6.1 Description

Oil compressibility describes the fractional change in oil volume with pressure. It is critical for material balance calculations and pressure transient analysis.

2.6.2 Explicit Method - EXPLT

$$c_o = -\frac{1}{B_o}\left[\frac{dB_o}{dP} - \frac{B_g}{5.61458}\frac{dR_s}{dP}\right] \tag{20}$$

Symbol Definitions:

- c_o: Oil compressibility (1/psi)
- B_o: Oil formation volume factor (rb/stb)
- B_g: Gas formation volume factor (rcf/scf)
- R_s: Solution gas-oil ratio (scf/stb)
- P: Pressure (psia)

Note: Derivatives are calculated numerically using finite differences.

2.7 API-Specific Gravity Conversions

2.7.1 API to Specific Gravity

$$\gamma_o = \frac{141.5}{\text{API} + 131.5} \tag{21}$$

2.7.2 Specific Gravity to API

$$\text{API} = \frac{141.5}{\gamma_o} - 131.5 \tag{22}$$

Symbol Definitions:

- **API**: Oil API gravity (degrees)
- γ_o: Oil specific gravity (water=1)

3 Gas PVT Calculations

3.1 Gas Z-Factor (Compressibility Factor)

3.1.1 Description

The Z-factor corrects the ideal gas law for real gas behavior. It is defined as:

$$PV = ZnRT \tag{23}$$

where Z = 1 for ideal gas, Z < 1 for most real gases at reservoir conditions.

3.1.2 Dranchuk-Abou-Kassem (1975) Method - DAK

$$Z = 1 + A_1\rho_r + A_2\rho_r^2 + A_3\rho_r^5 + A_4\rho_r^2(1 + A_5\rho_r^2)\exp(-A_5\rho_r^2) \tag{24}$$

$$+ \frac{A_6 + A_7\rho_r + A_8\rho_r^2}{T_{pr}^3}\rho_r^2 \tag{25}$$

where reduced density ρ_r is solved iteratively from:

$$\rho_r = \frac{0.27P_{pr}}{ZT_{pr}} \tag{26}$$

Symbol Definitions:

- Z: Gas compressibility factor (dimensionless)
- ρ_r: Reduced density (dimensionless)
- P_{pr}: Pseudo-reduced pressure $= P/P_c$
- T_{pr}: Pseudo-reduced temperature $= (T + 460)/T_c$
- P_c: Pseudo-critical pressure (psia)
- T_c: Pseudo-critical temperature (°R)
- $A_1, \ldots, A_8$: Correlation constants

3.1.3 Hall-Yarborough (1973) Method - HY

Implicit equation for reduced density y:

$$\frac{0.06125P_{pr}t}{y}\exp\left[-1.2(1-t)^2\right] = F(y) \tag{27}$$

where:

$$F(y) = -A_1 + A_2/y + A_3/y^2 + A_4/y^3 + A_5/y^4 + y^2(A_6 + A_7/y + A_8/y^2)(1 + A_9y^2)\exp(-A_9y^2) \tag{28}$$

Then Z is calculated from:

$$Z = \frac{A_1P_{pr}}{y} \tag{29}$$

where $t = 1/T_{pr}$ and $A_1, \ldots, A_9$ are correlation constants.

3.1.4 Burgoyne-Nielsen-Stanko (2025) Method - BUR

Universal EOS-based correlation for Z-factor, viscosity, and enthalpy. This is the most recent and comprehensive method, developed specifically to handle:

- High concentrations of non-hydrocarbons (CO_2, H_2S, N_2, H_2)
- Pure non-hydrocarbon streams (100% CO_2, 100% H_2)
- Hydrogen mixtures (only method supporting H_2)
- Full range of P-T conditions

Based on a tuned 5-component Peng-Robinson EOS model. The formulation methods are fully described in SPE-229932-MS, and the tuning datasets are available on GitHub. The correlation is implemented in pyResToolbox and provides open access to these calculations. **Reference:** SPE-229932-MS (ADIPEC 2025) - Burgoyne, M.W. (Santos Ltd), Nielsen, M.H., and Stanko, M. (Whitson AS)

Method Comparison:

- **DAK**: Best for hydrocarbon gases with low non-HC content
- **HY**: Fast alternative for standard conditions
- **WYW**: Modern update to classical methods
- **BUR**: Universal method, required for H_2 or high non-HC content

3.2 Pseudo-Critical Properties

3.2.1 Description

Pseudo-critical properties (temperature and pressure) are used to calculate the reduced properties needed for Z-factor correlations.

3.2.2 Piper-McCain-Corredor (1993) Method - PMC

Without Non-Hydrocarbons:

$$T_c = 169.2 + 349.5\gamma_g - 74.0\gamma_g^2 \quad (^\circ\text{R}) \tag{30}$$

$$P_c = 756.8 - 131.07\gamma_g - 3.6\gamma_g^2 \quad (\text{psia}) \tag{31}$$

With CO_2 and H_2S Corrections:

$$T_c' = T_c - \epsilon_T \tag{32}$$

$$P_c' = \frac{P_c - \epsilon_P}{1 - y_{CO_2} - y_{H_2S}} \tag{33}$$

where:

$$\epsilon_T = 120(y_{CO_2}^{0.9} - y_{CO_2}^{1.6}) + 15(y_{H_2S}^{0.5} - y_{H_2S}^{4}) \tag{34}$$

$$\epsilon_P = 440y_{CO_2} + 600y_{H_2S} - 170y_{CO_2}y_{H_2S} \tag{35}$$

Symbol Definitions:

- T_c: Pseudo-critical temperature (°R)
- P_c: Pseudo-critical pressure (psia)
- γ_g: Gas specific gravity (air=1)
- y_{CO_2}: CO_2 mole fraction
- y_{H_2S}: H_2S mole fraction
- y_{N_2}: N_2 mole fraction
- ϵ_T: Temperature correction
- ϵ_P: Pressure correction

3.2.3 Burgoyne-Nielsen-Stanko (2025) Methods - BUR / BNS

BUR and **BNS** are EOS-based critical property methods from SPE-229932-MS. They handle arbitrary mixtures of hydrocarbons with CO_2, H_2S, N_2, and H_2, including pure non-hydrocarbon streams.

- **BUR**: 5-component Peng-Robinson EOS with tuned binary interaction parameters
- **BNS**: Companion method with different mixing rule calibration

Both methods compute pseudo-critical properties directly from the EOS framework rather than using empirical correlations, making them suitable for:

- 100% CO_2 or high CO_2 streams
- Hydrogen mixtures (any concentration)
- Acid gas mixtures (high H_2S + CO_2)

Reference: SPE-229932-MS (ADIPEC 2025)

Method Comparison:

- **PMC**: Best for hydrocarbon gases with low non-HC content
- **SUT**: Classical Sutton (1985) correlation
- **BUR**: Universal EOS-based, high non-HC content
- **BNS**: Alternative EOS calibration, complements BUR

3.3 Gas Formation Volume Factor (B_g)

3.3.1 Description

The gas formation volume factor relates gas volume at reservoir conditions to its volume at standard conditions.

$$B_g = \frac{0.02827 \times Z \times (T + 460)}{P} \quad \text{(rcf/scf)} \tag{36}$$

Symbol Definitions:

- B_g: Gas formation volume factor (rcf/scf)
- Z: Gas compressibility factor (dimensionless)
- T: Temperature (°F)
- P: Pressure (psia)

3.4 Gas Viscosity (μ_g)

3.4.1 Description

Gas viscosity is required for flow calculations and increases with pressure and temperature.

3.4.2 Lee-Gonzalez-Eakin (1966) Method - LGE

$$\mu_g = K \times 10^{-4} \times \exp\left[X \times \rho_g^Y\right] \tag{37}$$

where:

$$K = \frac{(9.4 + 0.02M)(T + 460)^{1.5}}{209 + 19M + (T + 460)} \tag{38}$$

$$X = 3.5 + \frac{986}{T + 460} + 0.01M \tag{39}$$

$$Y = 2.4 - 0.2X \tag{40}$$

$$M = 28.97\gamma_g \tag{41}$$

Symbol Definitions:

- μ_g: Gas viscosity (cP)
- ρ_g: Gas density (lb/ft^3)
- M: Gas molecular weight (lb/lbmol)
- γ_g: Gas specific gravity (air=1)
- T: Temperature (°F)

3.5 Gas Density (ρ_g)

$$\rho_g = \frac{P \times 28.97 \times \gamma_g}{Z \times 10.73 \times (T + 460)} \quad \text{(lb/ft}^3\text{)} \tag{42}$$

Symbol Definitions:

- ρ_g: Gas density (lb/ft^3)

- P: Pressure (psia)
- γ_g: Gas specific gravity (air=1)
- Z: Gas compressibility factor (dimensionless)
- T: Temperature (°F)

3.6 Gas Compressibility (c_g)

3.6.1 Description

Gas compressibility describes the fractional change in gas volume with pressure.

$$c_g = \frac{1}{P} - \frac{1}{Z}\frac{dZ}{dP} \tag{43}$$

Symbol Definitions:

- c_g: Gas compressibility (1/psi)
- P: Pressure (psia)
- Z: Gas compressibility factor (dimensionless)

3.7 Gas Pseudopressure (ψ)

3.7.1 Description

Gas pseudopressure (or real gas potential) linearizes the gas diffusivity equation and is used in well test analysis.

$$\psi(P) = 2\int_{P_0}^{P} \frac{P'}{\mu_g Z} dP' \quad (\text{psia}^2/\text{cP}) \tag{44}$$

Symbol Definitions:

- ψ: Gas pseudopressure (psia^2/cP)
- P: Pressure (psia)
- P_0: Reference pressure (usually 0 psia)
- μ_g: Gas viscosity (cP)
- Z: Gas compressibility factor (dimensionless)

Note: Integral is evaluated numerically using trapezoidal rule or Simpson's rule.

4 Inflow Performance (Well Productivity)

4.1 Oil Rate - Radial Flow (Vertical Wells)

4.1.1 Description

Calculates oil production rate for vertical wells with radial flow geometry using Darcy's law for steady-state or pseudo-steady-state flow.

4.1.2 Darcy's Law - Undersaturated Reservoir

$$q_o = \frac{0.00708 \times k \times h \times (P_i - P_{wf})}{\mu_o \times B_o \times \left[\ln\left(\frac{r_e}{r_w}\right) - 0.75 + S\right]} \quad \text{(STB/day)} \tag{45}$$

Symbol Definitions:

- q_o: Oil production rate (STB/day)
- k: Permeability (mD)
- h: Net pay thickness (ft)
- P_i: Initial/reservoir pressure (psia)
- P_{wf}: Wellbore flowing pressure (psia)
- μ_o: Oil viscosity (cP)
- B_o: Oil formation volume factor (rb/stb)
- r_e: Drainage radius (ft)
- r_w: Wellbore radius (ft)
- S: Skin factor (dimensionless)

4.1.3 Vogel IPR - Saturated Reservoir

For saturated reservoirs ($P_i < P_b$), use Vogel's correlation:

$$\frac{q_o}{q_{o,max}} = 1 - 0.2\left(\frac{P_{wf}}{P_i}\right) - 0.8\left(\frac{P_{wf}}{P_i}\right)^2 \tag{46}$$

where maximum rate is:

$$q_{o,max} = q_o(P_{wf} = 0) \tag{47}$$

Additional Symbols:

- $q_{o,max}$: Maximum oil rate at zero bottomhole pressure (STB/day)
- P_b: Bubble point pressure (psia)

4.2 Oil Rate - Linear Flow (Horizontal Wells)

4.2.1 Description

Calculates oil production rate for horizontal wells or linear flow geometry.

$$q_o = \frac{0.00708 \times k \times h \times L \times (P_i - P_{wf})}{\mu_o \times B_o \times \left[\frac{r_e}{2} + S\right]} \quad \text{(STB/day)} \tag{48}$$

Additional Symbols:

- L: Horizontal well length (ft)

4.3 Gas Rate - Radial Flow (Vertical Wells)

4.3.1 Description

Calculates gas production rate for vertical wells using pseudopressure formulation.

$$q_g = \frac{0.00708 \times k \times h \times [\psi(P_i) - \psi(P_{wf})]}{\left[\ln\left(\frac{r_e}{r_w}\right) - 0.75 + S\right]} \quad \text{(MSCF/day)} \tag{49}$$

Symbol Definitions:

- q_g: Gas production rate (MSCF/day)
- $\psi(P)$: Gas pseudopressure (psia^2/cP)
- k: Permeability (mD)
- h: Net pay thickness (ft)
- P_i: Reservoir pressure (psia)
- P_{wf}: Wellbore flowing pressure (psia)
- r_e: Drainage radius (ft)
- r_w: Wellbore radius (ft)
- S: Skin factor (dimensionless)

4.4 Gas Rate - Linear Flow (Horizontal Wells)

$$q_g = \frac{0.00708 \times k \times h \times L \times [\psi(P_i) - \psi(P_{wf})]}{\left[\frac{r_e}{2} + S\right]} \quad \text{(MSCF/day)} \tag{50}$$

Additional Symbols:

- L: Horizontal well length (ft)

5 Relative Permeability Correlations

5.1 Corey (1954) Correlation

5.1.1 Description

Simple power-law model for relative permeability as a function of normalized saturation.

5.1.2 Normalized Saturation

Water Phase:

$$S_w^* = \frac{S_w - S_{wc}}{1 - S_{wc} - S_{or}} \tag{51}$$

Oil Phase (Water-Oil):

$$S_o^* = \frac{S_o - S_{or}}{1 - S_{wc} - S_{or}} \tag{52}$$

Gas Phase:

$$S_g^* = \frac{S_g - S_{gc}}{1 - S_{wc} - S_{org}} \tag{53}$$

Symbol Definitions:

- S_w^*: Normalized water saturation
- S_o^*: Normalized oil saturation
- S_g^*: Normalized gas saturation
- S_w: Water saturation
- S_o: Oil saturation
- S_g: Gas saturation
- S_{wc}: Connate water saturation
- S_{or}: Residual oil saturation (to water)
- S_{org}: Residual oil saturation (to gas)
- S_{gc}: Critical gas saturation

5.1.3 Relative Permeability Functions

Water Relative Permeability:

$$k_{rw} = k_{rw}^{max} \times (S_w^*)^{n_w} \tag{54}$$

Oil Relative Permeability (to water):

$$k_{row} = k_{row}^{max} \times (1 - S_w^*)^{n_o} \tag{55}$$

Gas Relative Permeability:

$$k_{rg} = k_{rg}^{max} \times (S_g^*)^{n_g} \tag{56}$$

Oil Relative Permeability (to gas):

$$k_{rog} = k_{rog}^{max} \times (1 - S_g^*)^{n_o} \tag{57}$$

Additional Symbols:

- k_{rw}: Water relative permeability
- k_{row}: Oil relative permeability (to water)
- k_{rg}: Gas relative permeability
- k_{rog}: Oil relative permeability (to gas)
- k_{rw}^{max}: Maximum water relative permeability
- k_{row}^{max}: Maximum oil relative permeability
- k_{rg}^{max}: Maximum gas relative permeability
- n_w: Corey exponent for water
- n_o: Corey exponent for oil
- n_g: Corey exponent for gas

Typical Values:

- n_w: 1.5 - 3.0
- n_o: 2.0 - 4.0
- n_g: 2.0 - 3.5

5.2 LET Correlation (Lomeland-Ebeltoft-Thomas 2005)

5.2.1 Description

Flexible three-parameter model (L, E, T) that can fit complex relative permeability curves.

Water Relative Permeability:

$$k_{rw} = k_{rw}^{max} \times \frac{(S_w^*)^{L_w}}{(S_w^*)^{L_w} + E_w(1 - S_w^*)^{T_w}} \tag{58}$$

Oil Relative Permeability:

$$k_{ro} = k_{ro}^{max} \times \frac{(1 - S_w^*)^{L_o}}{(1 - S_w^*)^{L_o} + E_o(S_w^*)^{T_o}} \tag{59}$$

Gas Relative Permeability:

$$k_{rg} = k_{rg}^{max} \times \frac{(S_g^*)^{L_g}}{(S_g^*)^{L_g} + E_g(1 - S_g^*)^{T_g}} \tag{60}$$

Symbol Definitions:

- L_w, L_o, L_g: Low saturation parameters (controls curve near $S^* = 0$)
- E_w, E_o, E_g: Elevation parameters (controls height/position)
- T_w, T_o, T_g: Top saturation parameters (controls curve near $S^* = 1$)

Typical Ranges:

- L: 1.0 - 3.0
- E: 1.0 - 2.0
- T: 1.0 - 3.0

Advantages over Corey:

- More flexible curve shapes
- Better fitting to experimental data
- Independent control of endpoint behavior
- Smooth transitions

5.3 Jerauld Correlation (2003)

5.3.1 Description

Two-parameter (a, b) model for relative permeability curves. Simpler than LET but more flexible than Corey, with asymmetric curve shapes.

Jerauld Relative Permeability:

$$k_r(S^*) = \frac{(S^*)^a}{(S^*)^a + b \cdot (1 - S^*)^a} \tag{61}$$

Normalized Saturation:

$$S^* = \frac{S_w - S_{wc}}{1 - S_{wc} - S_{or}} \tag{62}$$

Symbol Definitions:

- a: Shape parameter controlling curve curvature
- b: Asymmetry parameter controlling curve position
- S^*: Normalized saturation

Advantages:

- Only two parameters (simpler than LET's three per phase)
- Can produce asymmetric S-shaped curves
- Useful for fitting residual saturation zones

5.4 Relative Permeability Fitting

5.4.1 Description

pyResToolbox provides automated fitting of laboratory relative permeability data to Corey, LET, and Jerauld curve families.

Available Fitting Tools:

- `fit_relative_permeability` — fit data to a specified model (Corey, LET, or Jerauld)
- `fit_relative_permeability_best` — automatically compare all models and select the best fit
- `check_let_physical` — verify LET parameters produce physically reasonable curves
- `evaluate_jerauld` — evaluate Jerauld model at given saturations

Fitting Objective: Minimize sum of squared errors between measured k_r and model prediction:

$$\text{SSE} = \sum_{i=1}^{N} \left(k_{r,\text{measured}}(S_i) - k_{r,\text{model}}(S_i;\theta)\right)^2 \tag{63}$$

where θ represents the model parameters (n for Corey; L, E, T for LET; a, b for Jerauld).

6 Brine Properties

Important Note: pyResToolbox uses **different calculation methods** for CH_4-saturated versus CO_2-saturated brines. The methods are described separately below.

6.1 CH_4-Saturated Brine Properties

6.1.1 Description

For methane-saturated brines, pyResToolbox uses the Modified Spivey Correlation per McCain *Petroleum Reservoir Fluid Properties*, Chapter 4, page 160.

6.1.2 Calculation Method

General Approach:
Properties are calculated using empirical correlations that account for:

- Temperature and pressure effects
- Salinity (weight% NaCl, range 0-100%)
- Methane saturation fraction (0 = no methane, 1 = fully saturated)

Key Features:

- Density calculated from pure water density with salt and methane corrections
- Viscosity uses temperature-dependent base correlation with salinity adjustments
- Compressibility derived from density derivatives with respect to pressure
- Methane solubility from modified Henry's Law with salt correction (lambda and eta coefficients)
- Gas molar volume calculated using Z-factor correlation for methane
- Formation volume factor (Bw) accounts for molar volume changes due to dissolved methane

Properties Calculated:

- B_w: Water formation volume factor (rb/stb)
- ρ_w: Brine density (specific gravity, dimensionless)
- μ_w: Brine viscosity (cP)
- c_w: Brine compressibility (1/psi)
- R_w: Methane solubility (scf/stb)

Reference: McCain, W.D. (2011). *Petroleum Reservoir Fluid Property Correlations*, Chapter 4.

6.2 CO_2-Saturated Brine Properties

6.2.1 Description

For CO_2-saturated brines, pyResToolbox uses a thermodynamic approach with multiple specialized correlations that differ significantly from the methane case.

6.2.2 Calculation Methods

Phase Equilibrium:

- **Spycher & Pruess (2010)**: Modified Redlich-Kwong (SRK) Cubic Equation of State
- Provides CO_2-H_2O mutual solubility via activity coefficients
- Temperature-dependent EOS coefficients (different for T $<$ 99°C vs T $>$ 109°C)
- Blended calculations for 99-109°C transition range

Pure Brine Density:

- **Spivey et al.** correlation (modified)
- Reference: McCain *Petroleum Reservoir Fluid Property Correlations*, Chapter 4

CO_2-Corrected Brine Density:

- **Garcia (2001)** equation for CO_2 molar volume
- Combined with Spycher & Pruess x_{CO_2} and Spivey base density

Pure Brine Viscosity:

- **Mao-Duan (2009)** correlation

CO_2-Corrected Brine Viscosity:

- **Islam-Carlson (2012)** model
- Adjusts base viscosity for dissolved CO_2 mole fraction

Properties Calculated:

- x_{CO_2}: CO_2 mole fraction in brine
- y_{H_2O}: H_2O mole fraction in CO_2-rich phase
- ρ_{CO_2-gas}: CO_2-rich gas density (lb/ft^3)
- $\rho_{brine-sat}$: CO_2-saturated brine density (lb/ft^3)
- $\rho_{brine-pure}$: Pure brine density (lb/ft^3)
- $\mu_{brine-sat}$: CO_2-saturated brine viscosity (cP)
- $\mu_{brine-pure}$: Pure brine viscosity (cP)

- B_{w-sat}: CO_2-saturated brine FVF (rb/stb)
- R_w: CO_2 solubility (scf/stb)
- $c_{w-unsat}$: Undersaturated brine compressibility (1/psi)
- c_{w-sat}: Saturated brine compressibility (1/psi)

Salinity Input: Parts per million (ppm) NaCl
Key References:

- Spycher, N. and Pruess, K. (2010). Geochimica et Cosmochimica Acta
- Garcia, J.E. (2001). Fluid Phase Equilibria
- Mao, S. and Duan, Z. (2009). Chemical Geology
- Islam, A.W. and Carlson, E.S. (2012). Geothermics

6.3 Key Differences Between CH_4 and CO_2 Methods

Aspect	**CH_4-Saturated**	**CO_2-Saturated**
Phase Equilibrium	Henry's Law-based empirical	Cubic EOS (Spycher & Pruess)
Density Correction	Dissolved CH_4 molar volume	Garcia (2001) CO_2 molar volume
Viscosity Adjustment	Salinity effect factors	Islam-Carlson (2012) model
Salinity Units	Weight% NaCl (0-100)	ppm NaCl
Compressibility	From density derivatives	Numerical differentiation (sat/unsat)

Table 2: Comparison of CH_4 vs CO_2 Brine Calculation Methods

7 Reservoir Heterogeneity

7.1 Lorenz Coefficient

7.1.1 Description

The Lorenz coefficient quantifies reservoir heterogeneity based on the cumulative flow capacity vs cumulative storage capacity curve. It ranges from 0 (homogeneous) to 1 (completely heterogeneous).

$$L = 2 \times \int_0^1 [F(\phi) - \phi] d\phi \tag{64}$$

where $F(\phi)$ is the Lorenz curve:

$$F(\phi) = \frac{\text{Cumulative Flow Capacity}}{\text{Total Flow Capacity}} \tag{65}$$

Symbol Definitions:

- L: Lorenz coefficient (0-1)
- $F(\phi)$: Lorenz curve function
- ϕ: Cumulative storage fraction

Interpretation:

- $L = 0$: Perfectly homogeneous (all layers have same kh)
- $L = 0.3$: Low heterogeneity
- $L = 0.5$: Moderate heterogeneity
- $L = 0.7$: High heterogeneity
- $L \to 1$: Extreme heterogeneity

7.2 Dykstra-Parsons Coefficient (β)

7.2.1 Description

The Dykstra-Parsons coefficient measures permeability variation assuming a log-normal distribution.

$$\beta = \frac{k_{50} - k_{84.1}}{k_{50}} \tag{66}$$

where:

- k_{50} is the median permeability (50th percentile)
- $k_{84.1}$ is the 84.1th percentile permeability (1 standard deviation below mean in log space)

Symbol Definitions:

- β: Dykstra-Parsons coefficient (0-1)
- k_{50}: Median permeability (mD)
- $k_{84.1}$: 84.1th percentile permeability (mD)

Interpretation:

- $\beta < 0.5$: Low variation (homogeneous)
- $\beta = 0.5 - 0.7$: Moderate variation
- $\beta > 0.7$: High variation (heterogeneous)

7.3 Lorenz-Beta Conversion

7.3.1 Description

Conversion between Lorenz coefficient and Dykstra-Parsons beta parameter, assuming log-normal permeability distribution.

Beta to Lorenz:

$$L \approx 1 - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(-\frac{x^2}{2}\right) \times \exp(x\sigma) dx \tag{67}$$

where σ is related to β through:

$$\beta = 1 - \exp(-\sigma) \tag{68}$$

Approximate Conversion:

$$L \approx 0.632 \times \beta^{0.9} \tag{69}$$

Symbol Definitions:

- L: Lorenz coefficient
- β: Dykstra-Parsons coefficient
- σ: Standard deviation of $\ln(k)$

7.4 Layer Permeability Distribution

7.4.1 Description

Generates permeability distribution for layered reservoirs based on Lorenz coefficient.

Log-Normal Distribution:

$$k_i = k_{avg} \times \exp\left[\sigma \times \Phi^{-1}(p_i) - \frac{\sigma^2}{2}\right] \tag{70}$$

where:

- k_i is permeability of layer i
- k_{avg} is average permeability
- Φ^{-1} is inverse standard normal CDF
- $p_i = (i - 0.5)/N$ is cumulative probability
- N is number of layers

Symbol Definitions:

- k_i: Layer permeability (mD)
- k_{avg}: Average permeability (mD)
- σ: Standard deviation of $\ln(k)$
- Φ^{-1}: Inverse cumulative normal distribution
- N: Number of layers

8 Aquifer Models

8.1 Van Everdingen-Hurst Model

8.1.1 Description

The Van Everdingen-Hurst model calculates water influx from an aquifer into a reservoir. It uses dimensionless influence functions based on the radial diffusivity equation.

Water Influx:

$$W_e(t) = B \times \sum_{j=1}^{n} \Delta P_j \times W_D(t_D - t_{D,j}) \tag{71}$$

where:

$$B = 1.119 \times \phi \times c_t \times h \times r_e^2 \quad \text{(rb/psi)} \tag{72}$$

Dimensionless Time:

$$t_D = \frac{0.000264 \times k \times t}{\phi \times \mu \times c_t \times r_e^2} \tag{73}$$

Dimensionless Radius:

$$r_D = \frac{r_{aq}}{r_e} \tag{74}$$

Symbol Definitions:

- W_e: Cumulative water influx (rb)
- B: Aquifer constant (rb/psi)
- ΔP_j: Pressure drop at time step j (psia)
- W_D: Dimensionless water influx function
- t_D: Dimensionless time
- ϕ: Aquifer porosity (fraction)
- c_t: Total compressibility (1/psi)
- h: Aquifer thickness (ft)
- r_e: Reservoir radius (ft)
- r_{aq}: Aquifer outer radius (ft)
- k: Aquifer permeability (mD)
- μ: Water viscosity (cP)
- t: Time (days)

8.2 Influence Functions

The dimensionless water influx $W_D(t_D)$ is tabulated for various dimensionless radius ratios r_D. For:

- Infinite aquifer: $r_D = \infty$
- Finite aquifer: $r_D = 2, 3, 5, 10, etc.$

9 Geomechanics and Wellbore Stability

The geomechanics module provides 27 tools for stress analysis, wellbore stability, and reservoir geomechanics calculations. These are critical for drilling optimization, completion design, sand management, and induced seismicity assessment.

9.1 Vertical (Overburden) Stress

9.1.1 Description

The vertical stress at any depth is the weight of overlying rock and fluid column.

Onshore Wells:

$$\sigma_v = \frac{\bar{\rho}_{bulk} \times D}{144} \quad \text{(psia)} \tag{75}$$

Offshore Wells:

$$\sigma_v = \frac{\rho_w \times D_w + \bar{\rho}_{bulk} \times (D - D_w)}{144} \quad \text{(psia)} \tag{76}$$

Overburden Gradient:

$$\frac{d\sigma_v}{dD} = \frac{\bar{\rho}_{bulk}}{144} \quad \text{(psi/ft)} \tag{77}$$

Symbol Definitions:

- σ_v: Vertical stress (psia)
- $\bar{\rho}_{bulk}$: Average bulk density of overburden (lb/ft^3)
- D: True vertical depth below surface (ft)
- D_w: Water depth for offshore wells (ft)
- ρ_w: Water density (lb/ft^3) - 62.4 fresh, 64 seawater

Typical Values:

- Average bulk density: 140-160 lb/ft^3
- Overburden gradient: 0.95-1.1 psi/ft

9.2 Pore Pressure Prediction

9.2.1 Eaton Method (1975)

Sonic Method:

$$P_p = \sigma_v - (\sigma_v - P_{p,n}) \times \left(\frac{\Delta t_n}{\Delta t}\right)^n \tag{78}$$

Resistivity Method:

$$P_p = \sigma_v - (\sigma_v - P_{p,n}) \times \left(\frac{R}{R_n}\right)^n \tag{79}$$

Symbol Definitions:

- P_p: Pore pressure (psia)

- $P_{p,n}$: Normal pore pressure at depth (psia)
- Δt: Observed sonic transit time (μs/ft)
- Δt_n: Normal compaction trend sonic value (μs/ft)
- R: Observed resistivity (ohm-m)
- R_n: Normal compaction trend resistivity (ohm-m)
- n: Eaton exponent (3.0 for sonic, 1.2 for resistivity)

9.3 Effective Stress

9.3.1 Terzaghi and Biot Effective Stress

Terzaghi Effective Stress ($\alpha = 1$):

$$\sigma' = \sigma - P_p \tag{80}$$

Biot Effective Stress:

$$\sigma' = \sigma - \alpha P_p \tag{81}$$

where Biot coefficient:

$$\alpha = 1 - \frac{K_{bulk}}{K_{grain}} \approx 0.6 \text{ to } 1.0 \tag{82}$$

Symbol Definitions:

- σ': Effective stress (psi)
- σ: Total stress (psi)
- P_p: Pore pressure (psi)
- α: Biot coefficient (dimensionless)
- K_{bulk}: Rock bulk modulus (psi)
- K_{grain}: Grain (matrix) modulus (psi)

9.4 Horizontal Stress

9.4.1 Poroelastic Horizontal Stress Model

Minimum Horizontal Stress (Uniaxial Strain):

$$\sigma_h = \frac{\nu}{1-\nu}(\sigma_v - \alpha P_p) + \alpha P_p \tag{83}$$

With Tectonic Stress:

$$\sigma_h = \frac{\nu}{1-\nu}(\sigma_v - \alpha P_p) + \alpha P_p + \sigma_{tect} \tag{84}$$

Stress Regimes:

- Normal Faulting: $\sigma_v > \sigma_H > \sigma_h$
- Strike-Slip: $\sigma_H > \sigma_v > \sigma_h$
- Reverse Faulting: $\sigma_H > \sigma_h > \sigma_v$

Symbol Definitions:

- σ_h: Minimum horizontal stress (psi)
- σ_H: Maximum horizontal stress (psi)
- ν: Poisson's ratio (0.15-0.40 typical)
- σ_{tect}: Tectonic stress contribution (psi)

9.5 Elastic Moduli Relationships

9.5.1 Conversion Between Elastic Constants

Given Young's Modulus (E) and Poisson's Ratio (ν):

$$K = \frac{E}{3(1-2\nu)} \quad \text{(Bulk Modulus)} \tag{85}$$

$$G = \frac{E}{2(1+\nu)} \quad \text{(Shear Modulus)} \tag{86}$$

$$\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)} \quad \text{(Lamé's First Parameter)} \tag{87}$$

Plane Strain Modulus:

$$E' = \frac{E}{1-\nu^2} \tag{88}$$

Symbol Definitions:

- E: Young's modulus (psi)
- K: Bulk modulus (psi)
- G: Shear modulus (psi)

- λ: Lamé's first parameter (psi)
- ν: Poisson's ratio (dimensionless)
- E': Plane strain modulus (psi)

9.6 Rock Strength

9.6.1 Mohr-Coulomb Failure Criterion

Shear Failure Envelope:

$$\tau = c + \sigma_n \tan\phi \tag{89}$$

Principal Stress Form:

$$\sigma_1 = \text{UCS} + q\sigma_3 \tag{90}$$

where:

$$q = \frac{1+\sin\phi}{1-\sin\phi} = \tan^2\left(45 + \frac{\phi}{2}\right) \tag{91}$$

$$\text{UCS} = \frac{2c\cos\phi}{1-\sin\phi} \tag{92}$$

Symbol Definitions:

- τ: Shear stress at failure (psi)
- c: Cohesion (psi)
- σ_n: Normal stress on failure plane (psi)
- ϕ: Internal friction angle (degrees)
- σ_1: Maximum principal stress at failure (psi)
- σ_3: Minimum principal stress (confining stress) (psi)
- q: Mohr-Coulomb strength parameter
- **UCS**: Unconfined Compressive Strength (psi)

9.7 Fracture Gradient

9.7.1 Estimation Methods

Eaton Method (1969):

$$\frac{P_{frac}}{D} = \frac{\nu}{1-\nu}\left(\frac{\sigma_v - P_p}{D}\right) + \frac{P_p}{D} \tag{93}$$

From Known σ_h:

$$P_{frac} = \sigma_h \quad \text{(minimum case)} \tag{94}$$

Equivalent Mud Weight:

$$\text{EMW}_{frac} = \frac{P_{frac}}{0.052 \times D} \quad \text{(ppg)} \tag{95}$$

Symbol Definitions:

- P_{frac}: Fracture pressure (psi)
- $\mathbf{EMW}_{frac}$: Fracture equivalent mud weight (ppg)
- D: True vertical depth (ft)

9.8 Wellbore Stress (Kirsch Equations)

9.8.1 Stress at Wellbore Wall

Hoop (Tangential) Stress at Wall:

$$\sigma_\theta = \sigma_H + \sigma_h - 2(\sigma_H - \sigma_h)\cos 2\theta - P_w \tag{96}$$

Maximum Hoop Stress (at $\theta = 90$):

$$\sigma_{\theta,max} = 3\sigma_H - \sigma_h - P_w \tag{97}$$

Minimum Hoop Stress (at $\theta = 0$):

$$\sigma_{\theta,min} = 3\sigma_h - \sigma_H - P_w \tag{98}$$

Radial Stress at Wall:

$$\sigma_r = P_w \tag{99}$$

Symbol Definitions:

- σ_θ: Tangential (hoop) stress (psi)
- σ_r: Radial stress (psi)
- P_w: Wellbore pressure (mud pressure) (psi)
- θ: Angle from σ_H direction (degrees)

9.9 Breakout Width Calculation

9.9.1 Description

Borehole breakouts occur when hoop stress exceeds rock strength. The angular width indicates stress state.

Breakout Initiation Condition:

$$\sigma_\theta(\theta_b) = \text{UCS} + q \times \sigma_r \tag{100}$$

Breakout Width:

$$W_{bo} = \pi - 2\theta_b \quad \text{(radians)} \tag{101}$$

where θ_b is angle from σ_H where failure initiates.

9.10 Tensile Failure and Breakdown Pressure

9.10.1 Hydraulic Fracturing Initiation

Breakdown Pressure (Impermeable Rock):

$$P_{bd} = 3\sigma_h - \sigma_H + T_0 - P_p \tag{102}$$

Breakdown Pressure (Permeable Rock):

$$P_{bd} = \frac{3\sigma_h - \sigma_H + T_0 - \eta P_p}{1 - \eta} \tag{103}$$

where poroelastic constant:

$$\eta = \frac{\alpha(1 - 2\nu)}{1 - \nu} \tag{104}$$

Fracture Re-opening Pressure:

$$P_{reopen} = 3\sigma_h - \sigma_H - P_p \tag{105}$$

Propagation Pressure:

$$P_{prop} \approx \sigma_h \tag{106}$$

Symbol Definitions:

- P_{bd}: Formation breakdown pressure (psi)
- T_0: Tensile strength of rock (psi)
- η: Poroelastic constant (dimensionless)
- P_{reopen}: Fracture re-opening pressure (psi)
- P_{prop}: Fracture propagation pressure (psi)

9.11 Safe Mud Weight Window

Lower Bound (Kick Pressure):

$$\mathrm{MW}_{min} = \frac{P_p}{0.052 \times D} + \Delta_{OB} \quad \text{(ppg)} \tag{107}$$

Upper Bound (Lost Circulation):

$$\mathrm{MW}_{max} = \frac{P_{frac}}{0.052 \times D} - \Delta_{frac} \quad \text{(ppg)} \tag{108}$$

Collapse Bound:

$$\mathrm{MW}_{collapse} = \frac{P_{collapse}}{0.052 \times D} \quad \text{(ppg)} \tag{109}$$

Symbol Definitions:

- **MW**: Mud weight (ppg)

- Δ_{OB}: Overbalance safety margin (ppg)
- Δ_{frac}: Fracture safety margin (ppg)
- $P_{collapse}$: Collapse pressure from rock mechanics (psi)

9.12 Reservoir Compaction

9.12.1 Uniaxial Strain Compaction

Vertical Compaction:

$$\Delta h = C_m \times \Delta P_p \times h \tag{110}$$

Uniaxial Compaction Coefficient:

$$C_m = \frac{\alpha(1+\nu)(1-2\nu)}{E(1-\nu)} \tag{111}$$

Simplified Form:

$$\Delta h = \frac{\alpha \Delta P_p \times h}{M} \tag{112}$$

where constrained modulus:

$$M = \frac{E(1-\nu)}{(1+\nu)(1-2\nu)} \tag{113}$$

Symbol Definitions:

- Δh: Compaction (ft)
- h: Reservoir thickness (ft)
- ΔP_p: Pressure depletion (psi)
- C_m: Uniaxial compaction coefficient (1/psi)
- M: Constrained (P-wave) modulus (psi)

9.13 Stress Path During Depletion

9.13.1 Horizontal Stress Change with Pore Pressure

Stress Path Coefficient:

$$\gamma = \frac{\Delta \sigma_h}{\Delta P_p} = \alpha \frac{1-2\nu}{1-\nu} \tag{114}$$

Final Horizontal Stress:

$$\sigma_h^{final} = \sigma_h^{initial} + \gamma \times \Delta P_p \tag{115}$$

Effective Stress Change:

$$\Delta \sigma_h' = \Delta \sigma_h - \alpha \Delta P_p = (\gamma - \alpha) \Delta P_p \tag{116}$$

Symbol Definitions:

- γ: Stress path coefficient (typically 0.5-0.8)

- $\Delta\sigma_h$: Change in horizontal stress (psi)
- ΔP_p: Change in pore pressure (psi)

Note: For depletion ($\Delta P_p < 0$), horizontal stress decreases but effective stress increases.

9.14 Stress Polygon

9.14.1 Frictional Equilibrium Constraints

Maximum Stress Ratio (Frictional Limit):

$$\frac{\sigma_1'}{\sigma_3'} \leq \left(\sqrt{\mu^2+1}+\mu\right)^2 = q_f \tag{117}$$

Normal Faulting Constraint ($\sigma_v = \sigma_1$):

$$\sigma_h \geq \frac{\sigma_v - P_p}{q_f} + P_p \tag{118}$$

Reverse Faulting Constraint ($\sigma_v = \sigma_3$):

$$\sigma_H \leq (\sigma_v - P_p) \times q_f + P_p \tag{119}$$

Symbol Definitions:

- μ: Fault friction coefficient (0.6-0.85 typical, Byerlee's law)
- q_f: Frictional limit ratio

9.15 Fault Stability Analysis

9.15.1 Coulomb Failure Stress

Slip Tendency:

$$T_s = \frac{\tau}{\sigma_n'} \tag{120}$$

Dilation Tendency:

$$T_d = \frac{\sigma_1' - \sigma_n'}{\sigma_1' - \sigma_3'} \tag{121}$$

Coulomb Failure Stress:

$$\text{CFS} = \tau - \mu\sigma_n' - c \tag{122}$$

Slip Condition:

$$\text{CFS} \geq 0 \quad \text{or} \quad T_s \geq \mu \tag{123}$$

Critical Pore Pressure for Slip:

$$P_{p,crit} = \sigma_n - \frac{\tau - c}{\mu} \tag{124}$$

Symbol Definitions:

- T_s: Slip tendency (dimensionless)

- T_d: Dilation tendency (dimensionless)
- **CFS**: Coulomb Failure Stress (psi)
- τ: Shear stress on fault plane (psi)
- σ'_n: Effective normal stress on fault (psi)
- c: Fault cohesion (psi) - usually 0 for reactivation

9.16 Thermal Stress Effects

Thermal Stress (Plane Strain):

$$\sigma_{thermal} = -\frac{E\alpha_T \Delta T}{1-\nu} \tag{125}$$

Effect on Hoop Stress:

$$\Delta\sigma_\theta = \sigma_{thermal} \tag{126}$$

Implications:

- Cooling ($\Delta T < 0$): Tensile stress, reduces collapse risk, promotes fracturing
- Heating ($\Delta T > 0$): Compressive stress, increases collapse risk, inhibits fracturing

Symbol Definitions:

- $\sigma_{thermal}$: Thermal stress (psi)
- α_T: Linear thermal expansion coefficient (1/°F)
- ΔT: Temperature change (°F)

Typical Values:

- α_T: 5×10^{-6} to 8×10^{-6} 1/°F for rocks

9.17 UCS Estimation from Logs

9.17.1 Empirical Correlations

McNally (1987) - Sandstone:

$$\text{UCS} = 1200 \times \exp(-0.036 \times \Delta t) \quad \text{(psi)} \tag{127}$$

Horsrud (2001) - Shale:

$$\text{UCS} = 0.77 \times V_p^{2.93} \quad \text{(MPa, } V_p \text{ in km/s)} \tag{128}$$

Chang (2006) - General:

$$\text{UCS} = 2.28 + 4.1089 \times E \quad \text{(MPa, } E \text{ in GPa)} \tag{129}$$

Vernik (1993) - Carbonate:

$$\text{UCS} = 254 \times (1 - 2.7\phi)^2 \quad \text{(MPa)} \tag{130}$$

Symbol Definitions:

- Δt: Sonic transit time (μs/ft)
- V_p: P-wave velocity (km/s)
- ϕ: Porosity (fraction)

9.18 Critical Drawdown for Sand Production

Critical Flowing BHP:

$$P_{wf,crit} = \frac{3\sigma'_H - \sigma'_h + (1+q)P_p - \text{UCS}}{1+q} \tag{131}$$

Critical Drawdown:

$$\Delta P_{crit} = P_{res} - P_{wf,crit} \tag{132}$$

where $q = (1+\sin\phi)/(1-\sin\phi)$

Risk Assessment:

- Low risk: $\Delta P_{crit} > 1000$ psi, UCS > 2000 psi
- Moderate risk: ΔP_{crit} 500-1000 psi
- High risk: $\Delta P_{crit} < 500$ psi, UCS < 1000 psi

9.19 Multiple Failure Criteria Comparison

9.19.1 Available Criteria

1. Mohr-Coulomb (ignores σ_2):

$$\sigma_1 = \text{UCS} + q\sigma_3 \tag{133}$$

2. Drucker-Prager (smooth cone):

$$\sqrt{J_2} = k + \alpha I_1 \tag{134}$$

where $I_1 = \sigma_1 + \sigma_2 + \sigma_3$ and $J_2 = \frac{1}{6}[(\sigma_1-\sigma_2)^2 + (\sigma_2-\sigma_3)^2 + (\sigma_1-\sigma_3)^2]$

3. Mogi-Coulomb (includes σ_2 effect):

$$\tau_{oct} = a + b\sigma_{m,2} \tag{135}$$

where $\sigma_{m,2} = (\sigma_1+\sigma_3)/2$ and $\tau_{oct} = \frac{\sqrt{2}}{3}\sqrt{(\sigma_1-\sigma_2)^2 + (\sigma_2-\sigma_3)^2 + (\sigma_1-\sigma_3)^2}$

Note: Mohr-Coulomb is most conservative. True triaxial criteria (Mogi-Coulomb, Modified Lade) account for intermediate stress strengthening effect.

10 Phase Behavior

10.1 Rachford-Rice Flash Calculation

10.1.1 Description

The Rachford-Rice equation is used to calculate vapor and liquid phase compositions and amounts for a multi-component mixture at specified temperature and pressure.

Rachford-Rice Equation:

$$\sum_{i=1}^{n} \frac{z_i(K_i - 1)}{1 + V(K_i - 1)} = 0 \tag{136}$$

where:

- V is vapor mole fraction (solve iteratively)
- $L = 1 - V$ is liquid mole fraction

Component Mole Fractions:

$$x_i = \frac{z_i}{1 + V(K_i - 1)} \quad \text{(liquid)} \tag{137}$$

$$y_i = \frac{K_i \times z_i}{1 + V(K_i - 1)} \quad \text{(vapor)} \tag{138}$$

Symbol Definitions:

- z_i: Overall mole fraction of component i
- x_i: Liquid phase mole fraction of component i
- y_i: Vapor phase mole fraction of component i
- K_i: Equilibrium K-value for component i ($K_i = y_i/x_i$)
- V: Vapor phase mole fraction
- L: Liquid phase mole fraction
- n: Number of components

Solution Method: Newton-Raphson iteration:

$$V^{n+1} = V^n - \frac{f(V^n)}{f'(V^n)} \tag{139}$$

where:

$$f(V) = \sum_{i=1}^{n} \frac{z_i(K_i - 1)}{1 + V(K_i - 1)} \tag{140}$$

$$f'(V) = -\sum_{i=1}^{n} \frac{z_i(K_i - 1)^2}{[1 + V(K_i - 1)]^2} \tag{141}$$

11 Decline Curve Analysis (DCA)

11.1 Arps Decline Equations

11.1.1 Description

Arps (1945) empirical decline equations model production rate decline over time. Three regimes are defined by the decline exponent b.

Rate vs Time:

$$q(t) = q_i(1 + b \cdot D_i \cdot t)^{-1/b} \qquad \text{(Hyperbolic, } 0 < b < 1\text{)} \tag{142}$$

$$q(t) = q_i\, e^{-D_i \cdot t} \qquad \text{(Exponential, } b = 0\text{)} \tag{143}$$

$$q(t) = \frac{q_i}{1 + D_i \cdot t} \qquad \text{(Harmonic, } b = 1\text{)} \tag{144}$$

Cumulative Production:

$$N_p(t) = \frac{q_i^b}{D_i(1-b)} \left[q_i^{1-b} - q(t)^{1-b} \right] \qquad \text{(Hyperbolic)} \tag{145}$$

$$N_p(t) = \frac{q_i - q(t)}{D_i} \qquad \text{(Exponential)} \tag{146}$$

$$N_p(t) = \frac{q_i}{D_i} \ln\left(\frac{q_i}{q(t)}\right) \qquad \text{(Harmonic)} \tag{147}$$

Estimated Ultimate Recovery (EUR):

$$\text{EUR} = N_p(t_{abandon}) \quad \text{where } q(t_{abandon}) = q_{min} \tag{148}$$

Symbol Definitions:

- q_i: Initial production rate
- D_i: Initial decline rate (1/time)
- b: Decline exponent (0 = exponential, 0–1 = hyperbolic, 1 = harmonic)
- N_p: Cumulative production
- q_{min}: Economic limit rate for EUR calculation

11.2 Duong Decline Model

11.2.1 Description

Duong (2011) model for tight/unconventional reservoirs exhibiting long-duration transient flow.

Duong Rate-Time Relation:

$$q(t) = q_1 \cdot t(a, m) \tag{149}$$

$$t(a, m) = t^{-m} \exp\left[\frac{a}{1-m}(t^{1-m} - 1)\right] \tag{150}$$

Symbol Definitions:

- q_1: Rate at $t = 1$ day
- a: Intercept parameter
- m: Slope parameter

11.3 Available DCA Tools

- `fit_decline` — Fit Arps parameters to rate vs time data
- `fit_decline_cumulative` — Fit from cumulative production
- `decline_forecast` — Generate rate forecast
- `arps_rate` — Calculate rate at specified time
- `arps_cumulative` — Calculate cumulative at specified time
- `estimated_ultimate_recovery` — EUR from decline parameters
- `duong_rate` — Duong model rate calculation
- `fit_ratio` — Fit production ratio trends (WOR, GOR)
- `ratio_forecast` — Forecast production ratios

12 Material Balance

12.1 Gas Material Balance (P/Z Method)

12.1.1 Description

The P/Z plot provides a graphical and analytical method for estimating original gas in place (OGIP) by plotting P/Z vs cumulative gas production G_p.

Gas Material Balance Equation:

$$\frac{P}{Z} = \frac{P_i}{Z_i}\left(1 - \frac{G_p}{G}\right) \tag{151}$$

Linear Form:

$$\frac{P}{Z} = \frac{P_i}{Z_i} - \frac{P_i}{Z_i \cdot G} \cdot G_p \tag{152}$$

OGIP from Regression:

$$G = -\frac{\text{intercept}}{\text{slope}} = \frac{P_i/Z_i}{|m|} \tag{153}$$

Symbol Definitions:

- P: Reservoir pressure (psia | barsa)
- Z: Gas compressibility factor at P and T
- P_i: Initial reservoir pressure
- Z_i: Z-factor at initial conditions
- G_p: Cumulative gas production
- G: Original gas in place (OGIP)
- m: Regression slope

12.2 Oil Material Balance (Havlena-Odeh)

12.2.1 Description

The Havlena-Odeh formulation linearizes the general material balance equation for oil reservoirs, enabling graphical determination of OOIP.

General Oil Material Balance:

$$F = N[E_o + mE_g + E_{fw}] + W_e B_w \tag{154}$$

Underground Withdrawal (F):

$$F = N_p[B_o + (R_p - R_s)B_g] + W_p B_w - W_i B_w - G_i B_g \tag{155}$$

Oil Expansion (E_o):

$$E_o = (B_o - B_{oi}) + (R_{si} - R_s)B_g \tag{156}$$

Gas Cap Expansion (E_g):

$$E_g = B_{oi}\left(\frac{B_g}{B_{gi}} - 1\right) \tag{157}$$

Rock and Water Expansion (E_{fw}):

$$E_{fw} = \frac{B_{oi}(c_f + S_{wc}c_w)}{1 - S_{wc}}\Delta P \tag{158}$$

Symbol Definitions:

- N: Original oil in place (OOIP)
- N_p: Cumulative oil production
- R_p: Cumulative producing GOR
- m: Gas cap ratio $= GB_{gi}/(NB_{oi})$
- W_e: Cumulative water influx
- c_f: Formation compressibility
- c_w: Water compressibility
- S_{wc}: Connate water saturation

12.3 Available Material Balance Tools

- `gas_material_balance` — P/Z analysis with optional Cole plot and Havlena-Odeh regression
- `oil_material_balance` — Havlena-Odeh oil material balance for OOIP estimation

13 Nodal Analysis and VLP

13.1 Vertical Lift Performance (VLP)

13.1.1 Description

VLP correlations calculate flowing bottom hole pressure from tubing head pressure, accounting for gravity, friction, and acceleration pressure drops along the wellbore.

General Pressure Drop Equation:

$$\frac{dP}{dL} = \left(\frac{dP}{dL}\right)_{\text{gravity}} + \left(\frac{dP}{dL}\right)_{\text{friction}} + \left(\frac{dP}{dL}\right)_{\text{acceleration}} \tag{159}$$

Gravity Term:

$$\left(\frac{dP}{dL}\right)_g = \frac{\rho_m\, g\, \cos\theta}{g_c} \tag{160}$$

Friction Term:

$$\left(\frac{dP}{dL}\right)_f = \frac{f\, \rho_m\, v_m^2}{2\, g_c\, d} \tag{161}$$

Available VLP Correlations:

- **HB**: Hagedorn & Brown (1965) — most common for oil wells
- **WG**: Woldesemayat & Ghajar (2007) — modern, recommended
- **GRAY**: Gray (1974) — good for gas-condensate wells
- **BB**: Beggs & Brill (1973) — versatile, all inclinations

13.2 Operating Point

13.2.1 Description

The operating point is the intersection of the IPR (inflow) and VLP (outflow) curves, representing the stabilized production rate and flowing bottom hole pressure.

Operating Point Condition:

$$P_{wf,\text{IPR}}(q) = P_{wf,\text{VLP}}(q) \tag{162}$$

Solved numerically by finding the intersection of inflow and outflow curves.

13.3 VFP Tables

13.3.1 Description

VFP (Vertical Flow Performance) tables are pre-computed lookup tables of flowing BHP as a function of flow rate, THP, water fraction, gas fraction, and artificial lift quantity. Used by reservoir simulators (ECLIPSE, OPM).

13.4 Available Nodal Analysis Tools

- `flowing_bhp` — Flowing bottom hole pressure from THP and completion
- `ipr_curve` — Inflow performance relationship curve generation
- `outflow_curve` — VLP outflow curve generation
- `operating_point` — VLP/IPR intersection (stabilized rate and BHP)
- `generate_vfp_prod_table` — VFPPROD table for reservoir simulators
- `generate_vfp_inj_table` — VFPINJ table for injection wells

14 Sensitivity Analysis

14.1 Parameter Sweep

14.1.1 Description

Systematically varies one or more input parameters across a range of values and evaluates the effect on output quantities. Useful for understanding parameter sensitivity and optimization.

14.2 Tornado Analysis

14.2.1 Description

A tornado chart shows the relative impact of each parameter on a target output by varying each parameter independently between low and high values while holding others at their base case.

Sensitivity Index:

$$S_i = |f(x_i^{high}) - f(x_i^{low})| \tag{163}$$

Parameters are ranked by S_i to identify the most influential inputs.

14.3 Available Sensitivity Tools

- `parameter_sweep` — Run parameter sweeps across any tool
- `tornado_sensitivity` — Generate tornado sensitivity analysis

15 Brine VLE (Soreide-Whitson)

15.1 Soreide-Whitson VLE Model

15.1.1 Description

The Soreide & Whitson (1992) model uses a modified Peng-Robinson equation of state with special mixing rules for the water–hydrocarbon system, enabling calculation of gas solubility in brine, densities, viscosities, and formation volume factors at reservoir conditions.

The pyResToolbox implementation extends the original framework with:

- **Multi-gas support:** CO_2, H_2S, N_2, H_2 as explicit gas-phase mole fractions
- **Sechenov salting-out corrections** for each dissolved gas species
- **Calibrated per-gas viscosity corrections**
- **IAPWS-IF97 Region 1** freshwater density (34-term Gibbs function)
- **Spivey/McCain** salinity correction applied multiplicatively
- **Garcia apparent molar volumes** (Plyasunov) for dissolved gas density effects

Peng-Robinson EOS:

$$P = \frac{RT}{V-b} - \frac{a(T)}{V(V+b)+b(V-b)} \tag{164}$$

Fugacity Equilibrium:

$$f_i^L(T,P,\mathbf{x}) = f_i^V(T,P,\mathbf{y}) \quad \text{for each component } i \tag{165}$$

Key Outputs:

- Per-component solubility in brine (CO_2, H_2S, N_2, H_2, HC)
- Total solution GOR
- Brine and gas phase densities, viscosities, FVF
- Water content in gas phase
- Compressibility

15.2 Available Brine Tools

- `calculate_brine_properties` — CH_4 and CO_2 saturated brine properties
- `soreide_whitson_vle` — Full VLE brine-gas equilibrium calculation

16 Quick Reference Tables

16.1 Physical Constants

Constant	Symbol	Value
Universal gas constant	R	10.73 psia·ft^3/(lbmol·°R)
Standard pressure	P_{sc}	14.7 psia
Standard temperature	T_{sc}	60°F (520°R)
Air molecular weight	M_{air}	28.97 lb/lbmol
Water density at SC	ρ_w^{SC}	62.4 lb/ft^3
Conversion °F to °R	T_R	$T_F + 459.67$

Table 3: Physical Constants Used in Calculations

16.2 Conversion Factors

From	To	Multiply By
psia	bara	0.06895
°F	°C	$(T_F - 32) \times 5/9$
ft	m	0.3048
mD	m^2	9.869×10^{-16}
cP	Pa·s	0.001
STB	m^3	0.1590
MSCF	m^3	28.32
lb/ft^3	kg/m^3	16.02

Table 4: Common Unit Conversions

16.3 Typical Property Ranges

Property	Typical Range	Comments
API gravity	10 - 50°	Light crude: 30-50°, Heavy: 10-25°
Gas gravity	0.6 - 1.2	Dry gas: 0.6-0.7, Wet gas: 0.7-0.9
R_s at P_b	100 - 3000 scf/stb	Depends on oil type and pressure
B_o	1.01 - 2.0 rb/stb	Increases with R_s
μ_o	0.5 - 100 cP	Light oil: 0.5-5, Heavy: 10-100+
Z-factor	0.7 - 1.2	Lower at medium pressures
B_g	0.003 - 0.010 rcf/scf	Increases with T, decreases with P
μ_g	0.01 - 0.03 cP	Increases with P and T
Permeability	0.1 - 1000 mD	Tight: <1, Good: 100-1000
Porosity	0.05 - 0.35	Low: 0.05-0.15, Good: 0.20-0.30

Table 5: Typical Property Ranges for Petroleum Reservoirs

17 Correlation Method Recommendations

17.1 Oil PVT Methods

Property	Recommended	When to Use
Bubble Point	VALMC	Wide range of conditions, most accurate
	STAN	Classic correlation, quick estimates
	VELAR	Regional correlations, comparison
Solution GOR	VELAR	Default choice, good accuracy
	STAN	Compatibility with older studies
	VALMC	Alternative method
FVF	MCAIN	Most accurate, density-based
	STAN	Simple correlation, fast
Viscosity	BR	Industry standard, widely validated

Table 6: Recommended Correlations for Oil PVT

17.2 Gas PVT Methods

Property	Recommended	When to Use
Z-Factor	DAK	Most accurate, industry standard
	HY	Fast, good for most conditions
	WYW	Newer correlation, comparison
	BUR	High non-hydrocarbons, 100% CO_2, H_2
Critical Props	PMC	Best accuracy, accounts for non-HCs
	SUT	Classic method, compatibility
	BUR	High non-hydrocarbon content
	BNS	Alternative EOS calibration, complements BUR
Viscosity	LGE	Industry standard

Table 7: Recommended Correlations for Gas PVT

17.3 Relative Permeability Methods

Method	Use For	Advantages
Corey	Quick estimates Standard cases Initial models	Simple, fast, widely used Few parameters, easy to understand Good starting point
LET	History matching Complex systems Final models	Flexible curve shapes Better fit to lab data Independent endpoint control
Jerauld	Intermediate Asymmetric curves	Two parameters, simpler than LET Better than Corey for S-shaped curves

Table 8: Relative Permeability Method Selection

18 Tool Categories and Count

18.1 Available Tools by Category

18.2 Key Features

- **Production Ready**: All 108 tools tested and validated
- **Dual Units**: Field (psia, °F, ft) and Metric (barsa, °C, m)
- **Array Support**: Calculate properties at multiple pressures simultaneously
- **Multiple Methods**: Choose from various industry-standard correlations
- **Type Safety**: Pydantic models ensure input validation
- **Zero Configuration**: Works out of the box with Claude Desktop
- **GPL-3.0 Licensed**: Free and open source

19 Usage Notes and Best Practices

19.1 General Guidelines

19.1.1 Input Validation

- Always use absolute pressures (psia, not psig or barg)
- Temperature must be in Fahrenheit, not Celsius
- Check that input ranges are physically realistic
- Verify that saturation endpoints sum correctly (e.g., $S_{wc} + S_{or} < 1.0$)

19.1.2 Method Selection

- Use recommended methods unless specific requirements exist
- For bubble point: VALMC is most accurate for wide ranges
- For Z-factor: DAK is industry standard, BUR for high non-HCs
- For relative permeability: Corey for quick estimates, LET for accuracy

19.1.3 Common Mistakes to Avoid

- Using separator temperature instead of reservoir temperature
- Confusing R_{sb} (solution GOR at bubble point) with separator GOR
- Not setting Vogel=True when $P_i < P_b$ (saturated reservoir)
- Using wrong gas gravity (separator vs weighted average)
- Not accounting for non-hydrocarbon components (H_2S, CO_2, N_2)
- Forgetting to convert units from SI to Field Units

19.2 PVT Calculation Workflow

19.2.1 Typical Oil PVT Workflow

1. Calculate bubble point pressure (P_b) from R_{sb}, API, T, γ_g
2. Check if reservoir is undersaturated ($P_i > P_b$) or saturated ($P_i < P_b$)
3. Calculate $R_s(P)$ at desired pressures
4. Calculate $B_o(P, R_s)$ using McCain or Standing method
5. Calculate $\mu_o(P, R_s)$ using Beggs-Robinson
6. Calculate $\rho_o(B_o, R_s)$ using SWMH method
7. Calculate c_o from numerical derivatives of B_o
8. Generate complete PVT table for simulation

19.2.2 Typical Gas PVT Workflow

1. Calculate pseudo-critical properties (T_c, P_c) from γ_g and impurities
2. Calculate Z-factor at desired pressures using DAK or BUR
3. Calculate $B_g(P, T, Z)$ from real gas equation
4. Calculate gas density $\rho_g(P, T, Z)$
5. Calculate gas viscosity $\mu_g(\rho_g, T)$ using LGE method
6. Calculate gas compressibility c_g from Z-factor derivatives
7. Calculate pseudopressure $\psi(P)$ by integrating $P/(\mu_g Z)$

19.3 Well Performance Analysis

19.3.1 IPR Generation Steps

1. Gather reservoir data: P_i, P_b, API, T, h, k, S, r_e, r_w
2. Calculate PVT properties at average pressure
3. Choose flow geometry: radial (vertical well) or linear (horizontal well)
4. For saturated reservoir ($P_i < P_b$), set Vogel=True
5. Calculate rates at multiple flowing pressures: $P_{wf} = 0$ to P_i
6. Plot q vs P_{wf} to generate IPR curve
7. Identify AOF (absolute open flow = rate at $P_{wf} = 0$)

19.4 Simulation Preparation

19.4.1 Relative Permeability Table Generation

1. Choose table type: SWOF (water-oil), SGOF (gas-oil), or SGWFN (3-phase)
2. Choose correlation: Corey (simple) or LET (flexible)
3. Specify saturation endpoints: S_{wc}, S_{or}, S_{org}, S_{gc}
4. Set maximum relative permeabilities: k_{rw}^{max}, k_{ro}^{max}, k_{rg}^{max}
5. Define correlation parameters (Corey exponents or LET parameters)
6. Generate table with 20-50 saturation points
7. Validate endpoints and curves against core data
8. Export to ECLIPSE/CMG format for simulation

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21 Appendix: Nomenclature

21.1 Primary Symbols

Symbol	Units	Description
API	degrees	Oil API gravity
B_o	rb/stb	Oil formation volume factor
c_o	1/psi	Oil compressibility
μ_o	cP	Oil viscosity
ρ_o	lb/ft^3	Oil density
R_s	scf/stb	Solution gas-oil ratio
R_{sb}	scf/stb	Solution GOR at bubble point
P_b	psia	Bubble point pressure
γ_o	-	Oil specific gravity (water=1)
B_g	rcf/scf	Gas formation volume factor
c_g	1/psi	Gas compressibility
μ_g	cP	Gas viscosity
ρ_g	lb/ft^3	Gas density
Z	-	Gas compressibility factor (Z-factor)
γ_g	-	Gas specific gravity (air=1)
P_c	psia	Pseudo-critical pressure
T_c	°R	Pseudo-critical temperature
P_{pr}	-	Pseudo-reduced pressure
T_{pr}	-	Pseudo-reduced temperature
ψ	psia^2/cP	Gas pseudopressure
B_w	rb/stb	Water/brine formation volume factor
c_w	1/psi	Brine compressibility
μ_w	cP	Brine viscosity
ρ_w	lb/ft^3	Brine density
R_w	scf/stb	Gas solubility in brine
w_t	wt%	Salinity (weight percent NaCl)
k	mD	Permeability
h	ft	Pay thickness (net)
ϕ	fraction	Porosity
S_w	fraction	Water saturation
S_o	fraction	Oil saturation
S_g	fraction	Gas saturation
S_{wc}	fraction	Connate water saturation
S_{or}	fraction	Residual oil saturation
P	psia	Pressure
T	°F	Temperature
P_i	psia	Initial/reservoir pressure
P_{wf}	psia	Wellbore flowing pressure
q_o	STB/day	Oil production rate
q_g	MSCF/day	Gas production rate
r_e	ft	Drainage radius
r_w	ft	Wellbore radius
S	-	Skin factor

Symbol	Units	Description
L	ft	Horizontal well length
k_{rw}	-	Water relative permeability
k_{ro}	-	Oil relative permeability
k_{rg}	-	Gas relative permeability
S^*	-	Normalized saturation
L	-	Lorenz coefficient
β	-	Dykstra-Parsons coefficient
y_{CO_2}	fraction	CO_2 mole fraction
y_{H_2S}	fraction	H_2S mole fraction
y_{N_2}	fraction	N_2 mole fraction
y_{H_2}	fraction	H_2 (hydrogen) mole fraction
σ_v	psi	Vertical (overburden) stress
σ_H	psi	Maximum horizontal stress
σ_h	psi	Minimum horizontal stress
σ_θ	psi	Tangential (hoop) stress
σ_r	psi	Radial stress
σ'	psi	Effective stress
α	-	Biot coefficient
E	psi	Young's modulus
ν	-	Poisson's ratio
K	psi	Bulk modulus
G	psi	Shear modulus
c	psi	Cohesion
ϕ	degrees	Internal friction angle
UCS	psi	Unconfined compressive strength
T_0	psi	Tensile strength
τ	psi	Shear stress
σ_n	psi	Normal stress
μ	-	Fault friction coefficient
T_s	-	Slip tendency
T_d	-	Dilation tendency
CFS	psi	Coulomb Failure Stress
α_T	1/°F	Thermal expansion coefficient
ΔT	°F	Temperature change

Table 10: Nomenclature - Primary Symbols

21.2 Subscripts

22 Index of Formulas

22.1 By Property

Oil Properties

- Bubble Point Pressure (P_b): Standing (p.4), Valko-McCain (p.4), Velarde (p.4)
- Solution GOR (R_s): Standing (p.5), Valko-McCain, Velarde
- Formation Volume Factor (B_o): McCain (p.6), Standing (p.6)
- Viscosity (μ_o): Beggs-Robinson (p.7)
- Density (ρ_o): SWMH (p.7)
- Compressibility (c_o): Explicit method (p.8)
- API-SG Conversions (p.8)

Gas Properties

- Z-Factor: DAK (p.9), Hall-Yarborough (p.10)
- Critical Properties: PMC (p.10), Sutton, BUR
- Formation Volume Factor (B_g): Real gas equation (p.11)
- Viscosity (μ_g): Lee-Gonzalez-Eakin (p.11)
- Density (ρ_g): Ideal gas law corrected (p.11)
- Compressibility (c_g): From Z-factor (p.12)
- Pseudopressure (ψ): Integral formulation (p.12)

Well Performance

- Oil Rate - Radial: Darcy's law (p.13), Vogel IPR (p.13)
- Oil Rate - Linear: Darcy's law (p.14)
- Gas Rate - Radial: Pseudopressure formulation (p.14)
- Gas Rate - Linear: Pseudopressure formulation (p.14)

Relative Permeability

- Corey Correlation: Normalized saturation (p.15), Kr functions (p.15)
- LET Correlation: Three-parameter model (p.16)

Brine Properties

- Density (ρ_w): Pressure-temperature-salinity (p.17)
- Viscosity (μ_w): Salinity correction (p.17)
- Compressibility (c_w): From FVF derivative (p.18)

- Formation Volume Factor (B_w): Three-effect model (p.18)
- Gas Solubility (R_w): Henry's Law (p.18)

Heterogeneity

- Lorenz Coefficient: Integral formulation (p.19)
- Dykstra-Parsons Coefficient: Percentile definition (p.19)
- Lorenz-Beta Conversion: Approximate relations (p.20)
- Layer Distribution: Log-normal model (p.20)

Geomechanics

- Vertical Stress: Overburden calculation
- Pore Pressure: Eaton method (sonic/resistivity)
- Effective Stress: Terzaghi and Biot formulations
- Horizontal Stress: Poroelastic model, stress regimes
- Elastic Moduli: E, K, G, λ conversions
- Rock Strength: Mohr-Coulomb criterion
- Fracture Gradient: Eaton method, EMW conversion
- Wellbore Stress: Kirsch equations (hoop/radial)
- Breakout Width: Failure initiation angle
- Breakdown Pressure: Hydraulic fracturing initiation
- Mud Weight Window: Kick, collapse, lost circulation bounds
- Reservoir Compaction: Uniaxial strain model
- Stress Path: Depletion/injection effects
- Stress Polygon: Frictional equilibrium constraints
- Fault Stability: Coulomb failure stress, slip tendency
- Thermal Stress: Temperature-induced stress changes
- UCS from Logs: McNally, Horsrud, Chang, Vernik correlations
- Critical Drawdown: Sand production prediction
- Failure Criteria: Mohr-Coulomb, Drucker-Prager, Mogi-Coulomb

Decline Curve Analysis

- Arps Rate: Exponential, hyperbolic, harmonic decline
- Arps Cumulative: Integrated production

- EUR: Estimated ultimate recovery
- Duong: Tight/unconventional decline
- Ratio Analysis: WOR, GOR, WGR trends

Material Balance

- Gas P/Z: Linear regression for OGIP
- Oil Havlena-Odeh: Underground withdrawal vs expansion

Nodal Analysis / VLP

- Flowing BHP: HB, WG, GRAY, BB correlations
- IPR Curves: Inflow performance
- VLP Curves: Outflow performance
- Operating Point: IPR/VLP intersection
- VFPPROD/VFPINJ: Simulator lookup tables

Brine VLE

- Soreide-Whitson: VLE for brine-gas systems

Sensitivity

- Parameter Sweep: Multi-parameter variation
- Tornado Analysis: Ranked parameter impact

Other

- Van Everdingen-Hurst Aquifer: Water influx
- Rachford-Rice Flash: Phase equilibrium

End of Formula Reference Guide

Compiled by: Gabriel Serrão Seabra

For the latest version and updates, visit:
https://github.com/gabrielserrao/pyrestoolbox-mcp

Built on pyResToolbox:
Original library by Mark W. Burgoyne
https://github.com/mwburgoyne/pyResToolbox

Remember: Always verify formulas with original sources before professional use.

Category	Description	Tools
Oil PVT	Oil property calculations Bubble point, Rs, Bo, viscosity, density, PVT harmonization	19
Gas PVT	Gas property calculations Z-factor, critical props, Bg, viscosity, hydrate prediction	15
Inflow	Well performance Oil/gas rates for radial/linear flow	4
Simulation	Reservoir simulation support Rel perm (Corey, LET, Jerauld), aquifer, flash, PVTW	11
Nodal / VLP	Nodal analysis Flowing BHP, IPR/VLP curves, VFPPROD/VFPINJ	6
DCA	Decline curve analysis Arps decline, Duong, EUR, ratio analysis	9
Material Balance	Reserve estimation Gas P/Z and oil Havlena-Odeh	2
Brine	Brine properties CH_4/CO_2 brine, Soreide-Whitson VLE	3
Geomechanics	Wellbore stability & stress Stress analysis, breakouts, fracturing, sanding	27
Heterogeneity	Layer analysis Lorenz, beta, layer distributions	5
Recommend	Method selection Gas, oil, VLP method recommendations	4
Sensitivity	Parameter analysis Parameter sweeps, tornado charts	2
Library	Component properties Critical properties for 100+ components	1
Total		**108**

Table 9: pyResToolbox MCP Tool Categories

Subscript	**Meaning**
o	Oil phase
g	Gas phase
w	Water/brine phase
b	At bubble point
i	Initial conditions
wf	Wellbore flowing
c	Critical or connate
r	Relative or residual
pr	Pseudo-reduced
D	Dimensionless
sc	Standard conditions
ST	Stock tank conditions
sp	Separator conditions

Table 11: Common Subscripts